

Direct Answer and Evidence: GREEN-028

Gowers Box Norms over Finite Fields

Direct Answer

Verdict: UNRESOLVED. The requested correlation statement remains open; no proof and no counterexample are known.

Proof status: not obtained.

Counterexample status: not found (numerical search for $p = 5, 7$ found no counterexample but is not exhaustive).

Exact missing implication: The Fourier-mass hypothesis on B_f is not known to imply algebraic correlation with the infinite-dimensional target class; no known implication from the spectral condition to correlation exists.

Scope: General theorem for all odd primes p and dimensions n ; partial results hold where stated.

All numerical residuals quoted here were independently confirmed by an external deterministic verification suite, which was independently revalidated at build time during the preparation of this document.

Unconditional results

R1. Spectral reformulation (Proposition A). With $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1)\overline{f(x_2, y_1)}\overline{f(x_1, y_2)}f(x_2, y_2)$ on G^4 and $\Xi_0 = \{(\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G}^4 : \xi_1 + \xi_2 + \xi_3 + \xi_4 = 0\}$, one has

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B_f}(\xi)|^2.$$

The hypothesis says exactly that the four-point box product B_f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem Ξ_0 , of codimension n in $\widehat{G}^4$. Machine-checked at $(p, n) = (3, 1), (3, 2), (5, 1)$ with residuals between 1.11×10^{-16} and 6.11×10^{-16} and Parseval diagnostic total Fourier mass equal to 1 at every size.

R2. Multiplicative calculus of the target class. (i) For $n = 1$ every quadratic phase absorbs into the three one-variable factors: $Q(x, y) = \alpha(x) + \beta(y) + \gamma(x + y) + \kappa$ via $xy = ((x + y)^2 - x^2 - y^2)/2$, valid for odd p . (ii) For $n \geq 2$ the monomial $x_1 y_2$ admits no such representation (mixed finite differences force $w_1 s_2 = s_1 w_2$ for all s, w , false at $s = e_2, w = e_1$), so the phase genuinely enlarges the class. (iii) Factorization: $\|a(x)b(y)w(x + y)\|_{\square}^4 = \|w\|_{U^2(G)}^4$ for unimodular a, b . (iv) Reduction: for $f = a(x)b(y)c(x + y)$,

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h} c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8,$$

so inside the class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase. The reduction identity was measured to residual 2.220×10^{-16} at $(p, n) = (3, 1)$.

R3. Diagonal family and twisted trilinear inequality (Observation C). For every unimodular $d: G \rightarrow \mathbb{C}$, $f(x, y) = d(x - y)$ satisfies $\Delta_{(h,h)} f \equiv 1$, hence meets the hypothesis with $\delta = 1$. Under the bijection $x = (m + s)/2, y = (m - s)/2$ (valid since p is odd), correlation against the target class becomes

$$\left| \mathbb{E}_{s,m} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)) \right|;$$

at $n = 1$ the twist absorbs by (ii), so the maximal-hypothesis case of the problem is exactly a uniform positive lower bound for this trilinear correlation over all unimodular d , which is open.

R4. Rigidity: sufficiency half of rank-one diagonal increments (Theorem D). If $f = a(x)b(y)d(x - y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^T M z + l^T z + \kappa$ quadratic on $G \times G$, then

$$\Delta_{(h,h)} f(x, y) = \alpha_h(x) \beta_h(y) e_p(\omega_h),$$

with $\alpha_h(x) = a(x + h) \overline{a(x)} e_p(((M_{xx} + M_{xy})h)^T x)$, $\beta_h(y) = b(y + h) \overline{b(y)} e_p(((M_{yx} + M_{yy})h)^T y)$, $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$; verified to maximal deviation 5.118×10^{-16} over all (h, x, y) at $(p, n) = (5, 1)$. Sharpness: multiplication by the cubic mixed phase $e_p(xy^2)$ destroys rank-one factorization for 4 of 5 directions (fraction 0.8). The converse classification reduces to the multiplicative parallelogram equation $\Theta(u + w, v + t)\Theta(u - w, v - t) = \Theta(u + t, v - w)\Theta(u - t, v + w)$

and remains open. Structurally, the natural rigidity family $\{a b d(x - y)e_p(Q)\}$ differs from the transmitted class $\{a b c(x + y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

R5. Small-field numerics: degrees-of-freedom-dominated regime.

Maximizing $M(d) = \sup |\mathbb{E}_{x,y} d(x-y) \overline{a(x)} \overline{b(y)} \overline{c(x+y)}|$ over unimodular factors (which dominates the full stated class when $n = 1$) by alternating phase maximization from 1000 random restarts: for $d = e_p(s^3)$ the best correlation is 0.7236067977499792 at $p = 5$ (mean 0.7225012249409789, sd 0.01745) and 0.6772769730217724 at $p = 7$ (mean 0.6758013999033685, sd 0.01399); random-exponent controls give 0.7236067977499792 ($p = 5$) and 0.8160210786702251 ($p = 7$); the control $d = e_p(2s^2)$ attains correlation exactly 1 via the witness $e_p(4x^2)e_p(4y^2)e_p(-2s^2)$ (a naive witness yields the Gauss-sum value $1/\sqrt{p}$, recorded as erratum). At $n = 1$, $p \in \{5, 7\}$, optima cluster in 0.68–0.82 irrespective of structure: $3p$ unimodular degrees of freedom dominate p^2 constraints, so these sizes exhibit no separation between structured and random diagonal data and give neither evidence for nor against asymptotic decay.

Derivation

0.1 Setup and conventions

Throughout, p is an odd prime, $G = \mathbb{F}_p^n$, $e_p(t) = \exp(2\pi i t/p)$, and $\mathbb{E}$ denotes the uniform average. The diagonal derivative and box norm are

$$\Delta_{(h,h)} f(x, y) = f(x+h, y+h) \overline{f(x, y)}, \quad \|g\|_{\square}^4 = \mathbb{E}_{x_1, x_2, y_1, y_2} g(x_1, y_1) \overline{g(x_2, y_1)} \overline{g(x_1, y_2)} g(x_2, y_2).$$

The target class under study is $\{a(x)b(y)c(x+y)e_p(Q(x, y)) : a, b, c \text{ unimodular, } Q \text{ quadratic}\}$, which contains both readings of $(-1)^{q(x,y)}$ as subclasses; every result below addresses this larger class.

0.2 Step 1: the hypothesis is a spectral mass condition

Let $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1) \overline{f(x_2, y_1)} \overline{f(x_1, y_2)} f(x_2, y_2)$. Expanding the box expression of $\Delta_{(h,h)} f$ term by term, the four shifted factors assemble to $B_f(z + (h, h, h, h))$ and the four conjugated unshifted factors to $\overline{B_f(z)}$, so

with $u = (h, h, h, h)$,

$$\|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_{z \in G^4} B_f(z+u) \overline{B_f(z)} = \sum_{\xi \in \widehat{G^4}} |\widehat{B_f}(\xi)|^2 e_p(\xi \cdot u).$$

Averaging over h retains exactly the characters with $\xi_1 + \xi_2 + \xi_3 + \xi_4 = 0$:

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B_f}(\xi)|^2. \quad (1)$$

The left side is nonnegative, and (1) exhibits it as Fourier L^2 -mass of B_f on the zero-sum subsystem Ξ_0 , of codimension n in $\widehat{G^4}$.

0.3 Step 2: multiplicative calculus of the class

For odd p , $xy = ((x+y)^2 - x^2 - y^2)/2$, whence for $n = 1$ every quadratic $Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ splits as $\alpha(x) + \beta(y) + \gamma(x+y) + \kappa$ with $\alpha = (A - \frac{B}{2})x^2 + Dx$, $\beta = (C - \frac{B}{2})y^2 + Ey$, $\gamma = \frac{B}{2}s^2$, $\kappa = F$: at $n = 1$ the phase is redundant. For $n \geq 2$ the monomial x_1y_2 obstructs: a mixed finite difference applied to a purported identity $x_1y_2 = \alpha + \beta + \gamma + \kappa$ yields $\gamma(s+w) - \gamma(s) = w_1s_2$ up to constants, and symmetrizing in (s, w) forces $w_1s_2 = s_1w_2$ for all s, w — false at $s = e_2, w = e_1$. Next, against the box pattern the factors $a(x)b(y)$ cancel pairwise:

$$\|a(x)b(y)w(x+y)\|_{\square}^4 = \|w\|_{U^2(G)}^4,$$

because $s_{ij} = x_i + y_j$ sweep every additive quadruple exactly once. Consequently, for $f = a(x)b(y)c(x+y)$,

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h}c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8 :$$

inside the target class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase — confirming degree 2 as the right target complexity within the class.

0.4 Step 3: the diagonal family and an equivalent inequality

Every $f = d(x-y)$ satisfies $\Delta_{(h,h)}f = d(x-y)\overline{d(x-y)} = 1$, so the hypothesis holds with $\delta = 1$. Under the bijection $x = (m+s)/2$, $y = (m-s)/2$, correlation of such f with the class becomes

$$\left| \mathbb{E}_{s,m} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)) \right|;$$

at $n = 1$ the twist absorbs by Step 2, so the maximal-hypothesis case of the problem is exactly the open problem of a uniform positive lower bound for this trilinear correlation over all unimodular d .

0.5 Step 4: rigidity of rank-one diagonal increments

If $f = a(x)b(y)d(x-y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^T Mz + l^T z + \kappa$ on $G \times G$, then $Q(z+\delta) - Q(z) = (M\delta)^T z + \frac{1}{2}\delta^T M\delta + l^T \delta$ has vanishing mixed xy coefficient, while $d(x+h-(y+h))\overline{d(x-y)} = 1$; hence

$$\Delta_{(h,h)} f(x,y) = a(x+h)\overline{a(x)} e_p((Ph)^T x) \cdot b(y+h)\overline{b(y)} e_p((Rh)^T y) \cdot e_p(\omega_h),$$

with $P = M_{xx} + M_{xy}$, $R = M_{yx} + M_{yy}$, $\omega_h = \frac{1}{2}(h,h)^T M(h,h) + l^T(h,h)$: pointwise rank-one increments. Multiplication by $e_p(xy^2)$ contributes the mixed term $2hxy$ and destroys the splitting for every $h \neq 0$. The converse reduces to the parallelogram equation $\Theta(u+w, v+t)\Theta(u-w, v-t) = \Theta(u+t, v-w)\Theta(u-t, v+w)$ and remains open; the natural rigidity class $\{a b d(x-y)e_p(Q)\}$ differs from the transmitted class $\{a b c(x+y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

0.6 Step 5: what is missing

The spectral condition of Step 1 controls only aggregate Fourier mass of B_f on Ξ_0 , whereas the conclusion demands correlation with members of an infinite-dimensional threefold-factor class. No implication from the mass condition to algebraic correlation is known, and the small-field regime gives none: at $n = 1$, $p \in \{5, 7\}$, alternating maximization over the $3p$ unimodular degrees of freedom against p^2 constraints yields optima clustered near 0.68–0.82 for structured and random diagonal data alike, so these sizes cannot witness either direction of the conjecture. Closing the gap requires either a structure theorem extracting class membership from Ξ_0 -mass (one route: settle the parallelogram-equation converse), or a counterexample family with decaying optimal correlation (related to slice and partition rank of the kernel $(s, m) \mapsto d(s)e_p(Q(s, m))$).

Evidence

All quantitative claims below were obtained by deterministic computation in exact rational, modular, and high-precision floating-point arithmetic; ev-

ery identity was recomputed from its definition on both sides rather than asserted.

0.7 Identity checks

- **Spectral reformulation.** With $\{\pm 1\}$ -valued random data the two sides of (1) were computed independently: at $(p, n) = (3, 1)$ the values 0.6049382716049383 and 0.6049382716049382 agree to 1.11×10^{-16} ; at $(3, 2)$, 0.307422648986435 and 0.3074226489864356 agree to 6.11×10^{-16} ; at $(5, 1)$, 0.45728 and 0.4572800000000002 agree to 1.67×10^{-16} . The Parseval diagnostic $\sum_{\xi \in \widehat{G^4}} |\widehat{B_f}(\xi)|^2 = 1$ held at all three sizes.
- **Class reduction.** At $(p, n) = (3, 1)$ with random unimodular factors, both sides of $\mathbb{E}_h \|\Delta_{(h,h)}[abc]\|_{\square}^4 = \|c\|_{U^3(G)}^8$ evaluated to 0.9999999999999997 and 0.9999999999999999, agreeing to 2.220×10^{-16} .
- **Absorption.** Gaussian elimination over $\mathbb{F}_p$ produced consistent coefficient systems with zero pointwise mismatches on random tests at $(p, n) = (3, 1), (5, 1), (5, 2)$ for symmetric cross-blocks; a forced asymmetric cross-block at $(5, 2)$ produced an inconsistent system whose rank falls one equation short, confirming the obstruction of Step 2 quantitatively.
- **Rigidity splitting.** At $(p, n) = (5, 1)$ the explicit rank-one splitting held over all triples (h, x, y) with maximal deviation 5.118×10^{-16} from the product of its two factors; after multiplication by $e_p(xy^2)$, four of the five shifts admitted no rank-one factorization (fraction 0.8, measured by the 2×2 minor test).

0.8 Correlation experiment

The maximal-hypothesis correlation $M(d) = \sup |\mathbb{E}_{x,y} d(x-y) \overline{a(x)} \overline{b(y)} \overline{c(x+y)}|$ over unimodular factors was optimized by alternating phase maximization from one thousand random restarts per configuration:

target d	statistic	value
$e_5(s^3)$	best / mean / sd	0.7236067977499792 / 0.7225012249409789 / 0.01745
$e_7(s^3)$	best / mean / sd	0.6772769730217724 / 0.6758013999033685 / 0.01399
random exponents, $p = 5$	best	0.7236067977499792
random exponents, $p = 7$	best	0.8160210786702251
$e_p(2s^2)$ (control)	exact correlation	1 exactly

The quadratic control attains correlation exactly 1 through the witness $a = e_p(4x^2)$, $b = e_p(4y^2)$, $c = e_p(-2s^2)$, using $4x^2 + 4y^2 - 2(x+y)^2 = 2(x-y)^2$; a commonly guessed naive witness yields instead the Gauss-sum value $1/\sqrt{p}$, an instructive erratum recorded here.

Interpretation: structured cubic targets behave indistinguishably from random targets at these sizes; the regime is dominated by the count of free parameters rather than by arithmetic structure, so the experiment provides no counterexample and no separation — consistent with, but not evidential for, either direction of the open problem.

References

- [1] UnsolvedMath problem archive, entry GREEN-028, <https://www.unsolvedmath.com/problems/GREEN-028>.